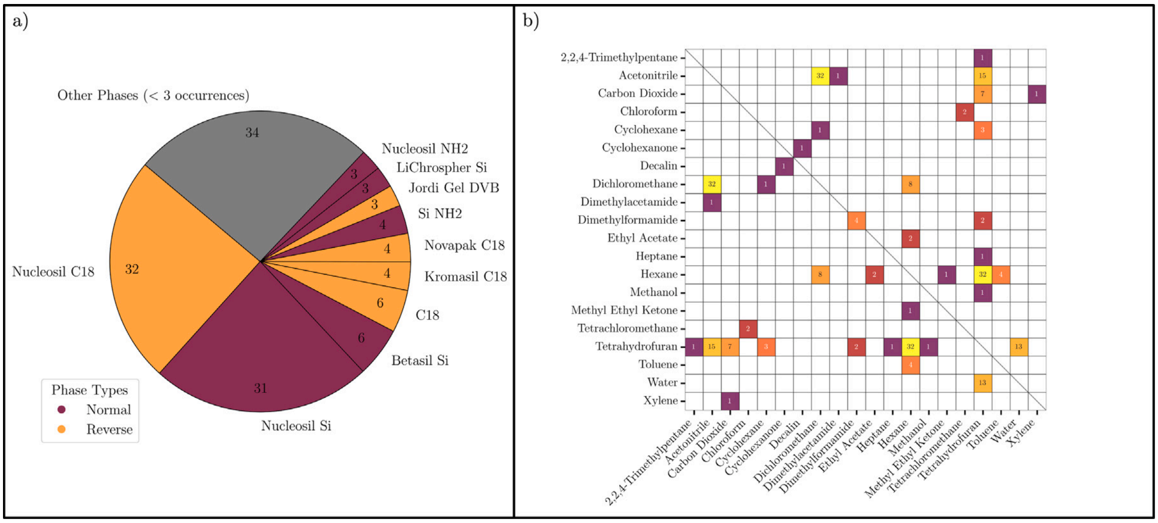

Fig. 6. (a) Pie chart of stationary phases reported used for Polystyrene, colored by Stationary Phase Type, and (b) heatmap of the solvent pairings used for Polystyrene. Numbers in each solvent pair indicate number of entries present in *PolyCrit*. Diagonal line highlights the mirrored quality of the heatmap.

Table 3
The five most common solvent solutions for Polystyrene entries with normal and reverse-phase columns. A (P) or (NP) indicates a solvent as either polar or nonpolar, respectively.

Normal phase	
Solvent solutions	**Occurrences**
Hexane (NP), THF (P)	35
MEK (P), Cyclohexane (NP)	10
DCM (P), Hexane (NP)	9
TCM (NP), Chloroform (NP)	9
THF (P), Cyclohexane (NP)	8
Reverse phase	
Solvent solutions	**Occurrences**
THF (P), Water (P)	38
DCM (P), ACN (P)	33
ACN (P), THF (P)	23
Acetone (P), Water (P)	19
Water (P), ACN (P)	17

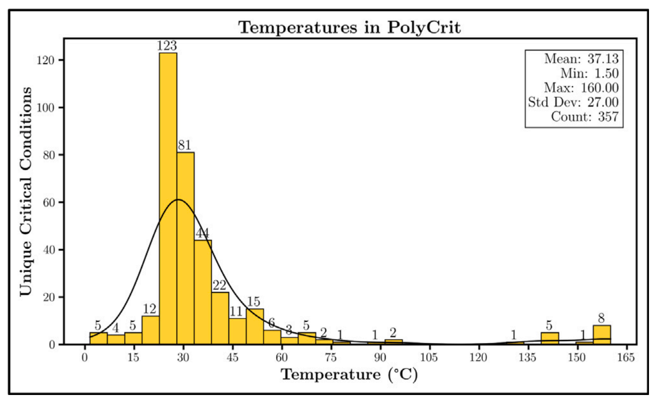

Fig. 7. Histogram of temperature with a box of key statistics. Black curves show the probability density function.

— typically porous silica beads — whereas in Reverse phases, the substrate is hydrophobic, with C18-terminated silica being a common example. The choice of solvent is closely tied to the phase type, users are encouraged to visit the website to explore specific solvent pairings captured in the database, which can aid in selecting the most commonly used or effective solvents for their desired stationary phase type and polymer.

To give a sample of the critical conditions for the most common polymer Polystyrene, Fig. 6 provides both the stationary phases (Fig. 6a) and the solvent pairs (Fig. 6b). The large proportion of Nucleosil Si/C18 stationary phases is a commonality among many of the other critical conditions in *PolyCrit*. Notice in Fig. 6b the small portion of critical solutions that take a single solvent (Dimethylformamide) instead of a pairing. Critical solutions containing three solvents are necessarily excluded from Fig. 6b by the nature of a heat map but can be found on *PolyCrit*.

Additionally, since as many as 33 parameters were tracked from each chromatographic entry, we are able to track the name, phase, and end group modifications of each stationary phase present in *PolyCrit*. In Table 3, we present the five most common solvent solutions used for the two main stationary phases (Normal and Reverse) of the columns. Our review of the data confirms that it is common practice to pair normal-phase columns with nonpolar solvents and reverse-phase columns with polar solvents—a standard approach in the field.

The last two figures (Figs. 7 and 8) display histograms of numeric-based parameters. Specifically, Fig. 7 gives an enlarged view (as compared to the smaller histograms of Fig. 8) of how temperature ranges across the critical conditions in *PolyCrit*. Note that the number of recorded temperature values (357) is different from the total entry count of 428; entries without a reported temperature have this parameter left blank. This approach applies to all parameters in the database; blank parameters were not replaced with default values. Naturally, the majority of temperatures lies close to room temperature (25 °C), but several studies can be found that were conducted with high temperature columns (> 100 °C).

In Fig. 8, all the numeric-based parameters are shown. Some histograms (marked with an asterisk) are shorted to only present the bottom 95th percentile of the data in order to remove high-valued outliers. However, such information is still preserved in the maximums recorded in the key statistics boxes.

Statistics presented in this section serve to give the reader a quick reference to information about the database, but many other comparisons and statistics can be made by the curious reader from the many entries present in *PolyCrit*.

5. Conclusion

The development of this database marks a significant advancement in the field of polymer research, with the potential to impact both